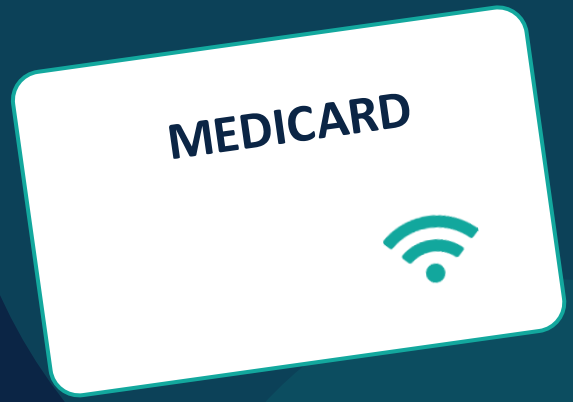
Medi Card

AI INTEGRATED SMART HEALTH CARD SYSTEM FOR ADVANCED PATIENT MEDICAL RECORD SECURITY

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Problem Statement & Proposed Solution

Problem Statement

Medical records are often maintained through paper-based documents, making them difficult to access, update, and carry when needed. Patients may also face difficulties when claiming **medical insurance**, as they need to provide previous medical records, prescriptions, and treatment details. This can lead to delays in treatment and insurance processing. **Medi Card** aims to provide a secure digital system that links a patient's unique ID with their medical records, allowing authorized users to quickly access the required information when needed.

Proposed Solution

MediCard is a secure digital healthcare solution that uses **NFC technology** to provide quick access to a patient's medical records through a unique Patient ID. The system reduces dependency on paper-based records and helps healthcare professionals access important medical information when required. It can also support **medical insurance claim processing** by making relevant patient records easier to retrieve. NFC is well suited for this application because it is contactless, fast, low-cost, battery-free, and simple to operate. Its affordability and minimal infrastructure requirements make it suitable for hospitals, clinics, and rural healthcare centers across India.

Objectives

Medi Card aims to improve the healthcare system by providing **quick, secure, and organized access to patient medical records** through NFC technology. It reduces dependence on paper documents, helps doctors make informed decisions using previous medical information, and simplifies the sharing of records when required for **medical insurance**. The system is especially beneficial for **elderly people**, who may find it difficult to carry and manage multiple prescriptions and medical documents. By providing their essential medical information through a simple NFC card, MediCard can make healthcare access more convenient, faster, and more reliable.



Quick & Easy Access
Enable fast retrieval of medical information using NFC technology.



Secure & Private
Ensure patient data is stored securely and accessible only to authorized users.



Centralized Storage
Store patient records in Firebase for real-time access and management.



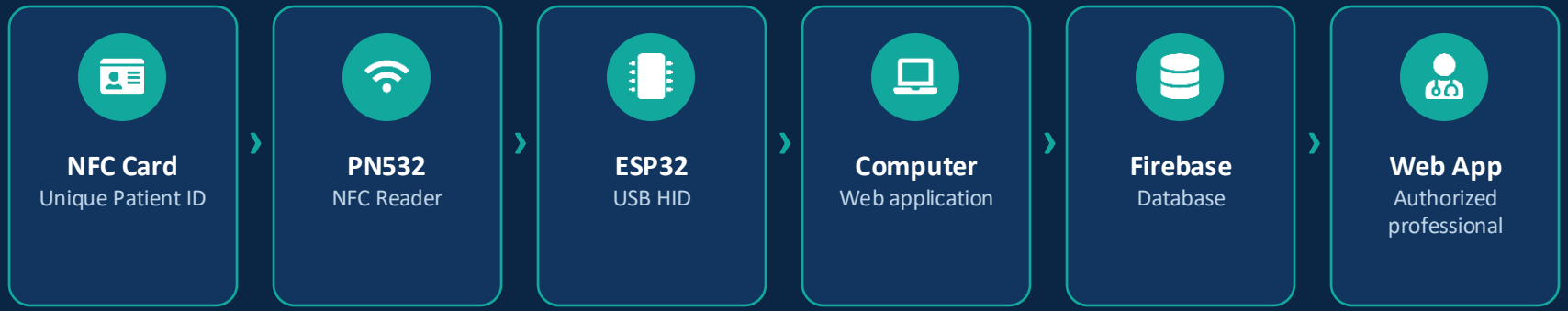
Support Healthcare Professionals
Help doctors view accurate patient history for better decision-making.



Reduce Paperwork
Minimize dependence on paper records and simplify insurance claim processes.

Working Principle

Medi Card works by using an **NFC card containing a unique Patient ID**. When the card is placed near the **PN532 NFC reader**, the ID is read and sent to the **ESP32 or Arduino uno**. The ESP32 transfers this ID to the computer through **USB HID (HID – Human Interface Device).** The web application then uses the Patient ID to search the **Firebase database** and retrieves the corresponding medical records. The authorized healthcare professional can then view the patient's information through the MediCard web application.



Hardware & Software Components

Hardware Components


NFC Card


PN532 NFC Reader


ESP32


USB connection


Computer/Laptop

Software Components


ESP32/Arduino uno firmware


NFC communication

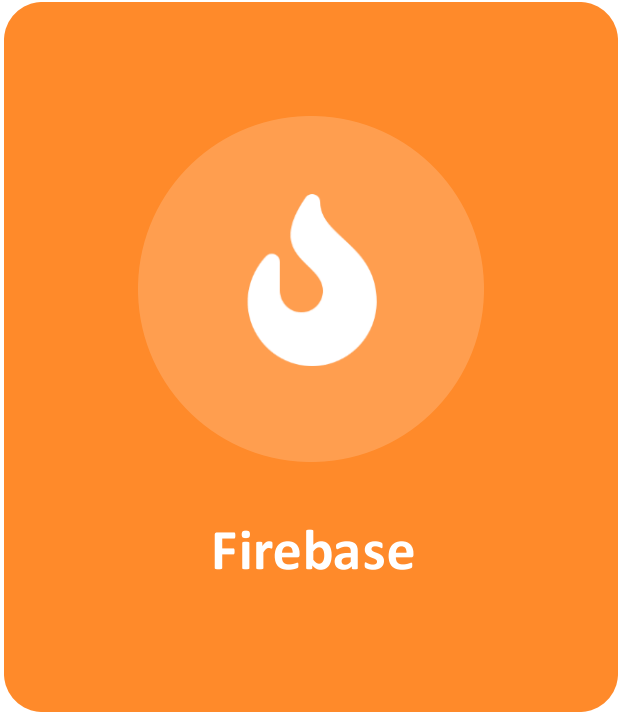

USB HID communication


Firebase Database


Web-based Medi Card application

Firebase

Firebase acts as the central database for the MediCard system, where patient information is stored and managed digitally. Each patient is assigned a unique Patient ID, which is linked to their medical records. When the NFC card is scanned, the Patient ID is sent to the web application, which searches Firebase for the matching record. The required information can then be retrieved and displayed to an authorized healthcare professional. Firebase also allows records to be updated and accessed in real time, making it suitable for a connected healthcare system.



Google AI Studio and Bolt in MediCard

Google AI Studio

Can be used to develop and test the **AI-related features** of Medi Card, such as designing prompts, processing medical information, and experimenting with AI-assisted functions.

Bolt

Used to build the **Medi Card web application frontend**. It provides the interface where authorized healthcare professionals can enter or receive the Patient ID and view the corresponding medical records retrieved from Firebase.

Flow:

NFC Card → ESP32 → Computer → Bolt Web App → Firebase → Patient Records

Google AI Studio → AI features / intelligent assistance

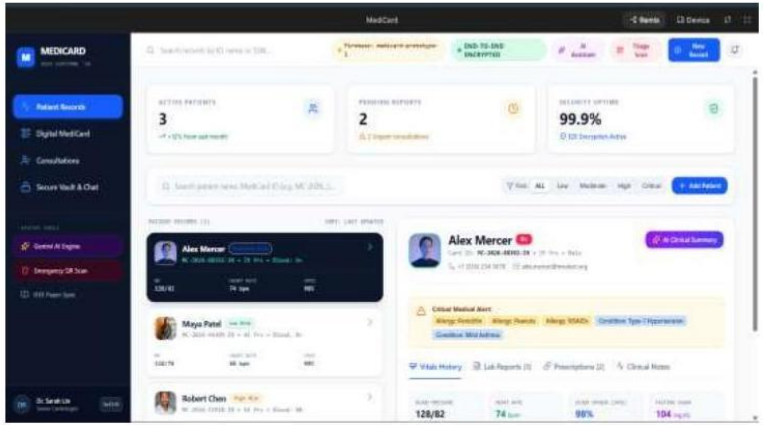
Security

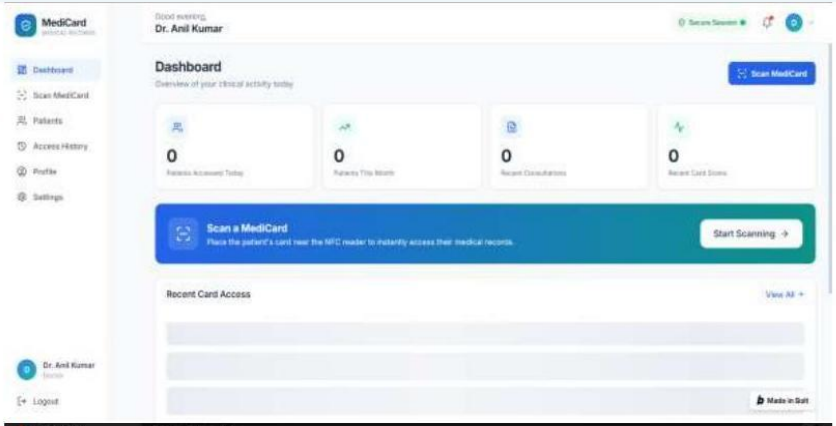
Medi Card is designed with security in mind by storing only a **unique Patient ID** on the NFC card rather than the complete medical record. The actual patient information is stored securely in **Firebase** and can be accessed only through authorized users. Authentication and access control help prevent unauthorized access, while secure communication protects data during transfer. This approach helps maintain **patient privacy, data integrity, and confidentiality**.

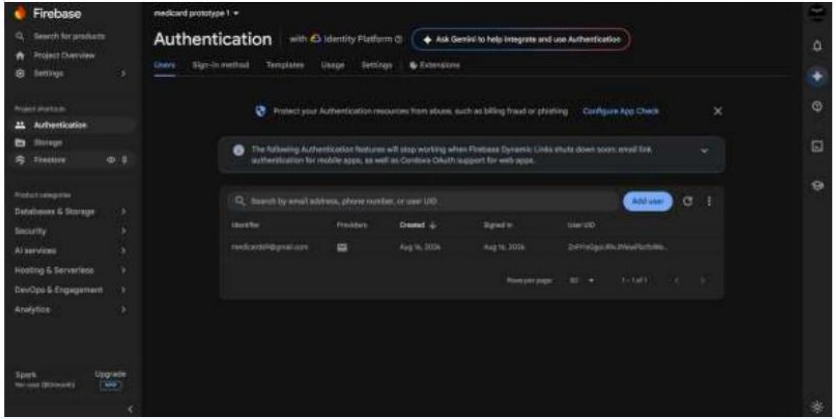


PROTOTYPE

PROTOTYPE







Advantages & Future Scope

Advantages

MediCard provides **quick and easy access to medical records** while reducing dependence on paper documents. It helps doctors view relevant patient history, saves time during consultations, and can simplify the retrieval of information for **medical insurance claims**. The NFC-based system is simple, contactless, low-cost, and convenient, making it suitable for **elderly people and healthcare centers with limited infrastructure**. It also provides a scalable foundation for connecting hospitals, clinics, laboratories, and other healthcare services.

Future Scope

MediCard can be further developed into a **unified digital healthcare platform** connecting hospitals, clinics, laboratories, pharmacies, and medical insurance services. Future versions can include a **mobile application, stronger authentication, encrypted communication, emergency medical information, and AI-assisted healthcare features**. The system can also support secure sharing of medical records between authorized healthcare providers, making it more useful for elderly patients and people who frequently visit different healthcare facilities.

THANK YOU

